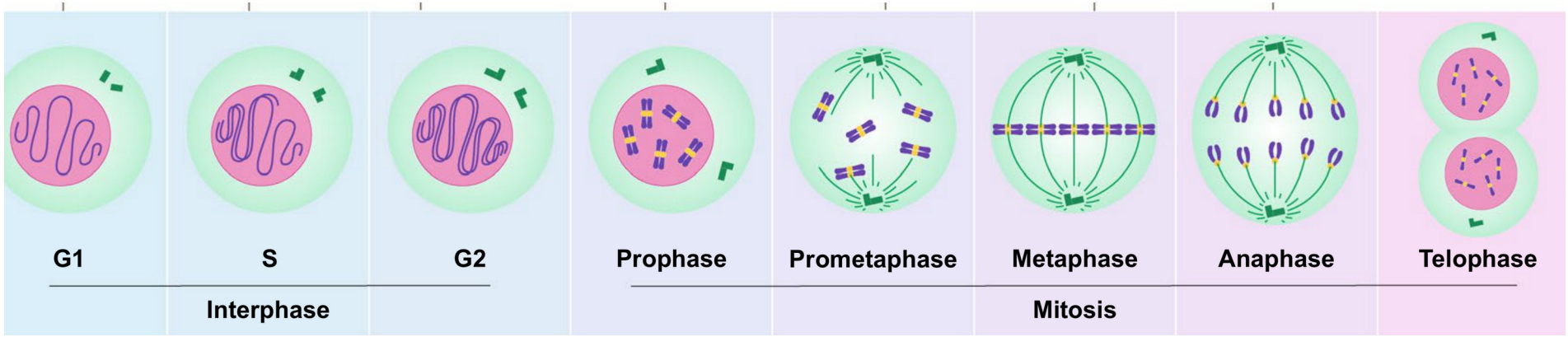
Rebalancing the kinetochore

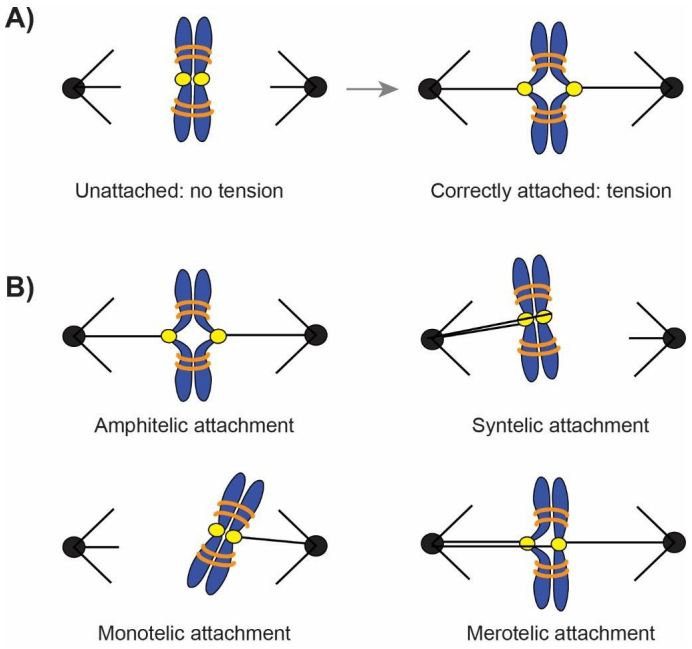
A drug-repurposing case for one child with mosaic variegated aneuploidy

“Every cell-cycle drug in clinical use was built to break the cell cycle. Not one was built to make it work better.”

MVA Hackathon 2026 · Track 2 · proband WGS_EX2312012 · BUB1B p.Leu737Ter / p.Asn1002Lys

THE PROBLEM: every cell division is a chance to get the chromosome count wrong





A checkpoint guards the hand-off

Each chromosome must grip both poles. Three of the four ways it can attach are wrong, so the cell holds mitosis until every one is right.

In this child the guard is weak

MVA: cells are born with the wrong chromosome count, giving growth failure, microcephaly and childhood cancer risk.

Bloom & North 2021 PMID 33892776 · McVey 2021 PMID 34445523, CC BY 4.0

THE QUESTIONS we made this work answer, before we let ourselves pick anything

1 What do these two alleles actually leave behind?

Not “pathogenic”. How much protein, and with which domains.

2 Is the deficit reversible at all?

If restoring BubR1 does nothing, no drug can work. Someone tested this.

3 Is there any drug that raises BubR1, or its phosphatase?

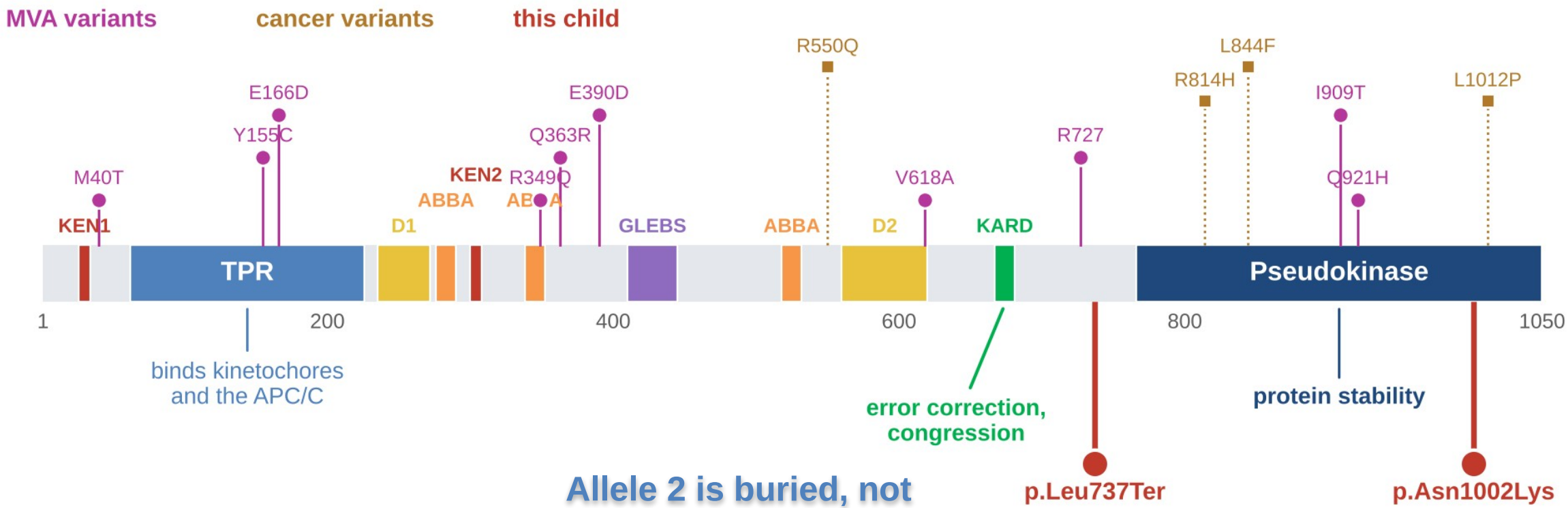
Asked across the whole druggable universe, not a shortlist.

4 If not, what else sits on the same node — and at what dose?

Answered on measured human exposure, or not answered.

Each question had to close before the next one opened

Q1 — Two alleles, and the domain that sits between them



Allele 1 truncates at 737

747 nt upstream of the last exon-exon junction, far past the 50-55 nt rule, so the transcript is destroyed.

Allele 2 is buried, not surface

87% buried, so PREDICTED to be made then degraded: a dosage mutant rather than an interface one. Abundance in patient cells is not yet measured.

Both land on the KARD

667–681: the motif that recruits the phosphatase PP2A-B56.

Domain layout adapted from Bloom & North 2021, PMID 33892776

SCALABILITY (15%) — we scored every BUB1B allele, not this child's two

Scalability 15%

Every allele, scored the same way

2,536 alleles
atlas + 854 full texts

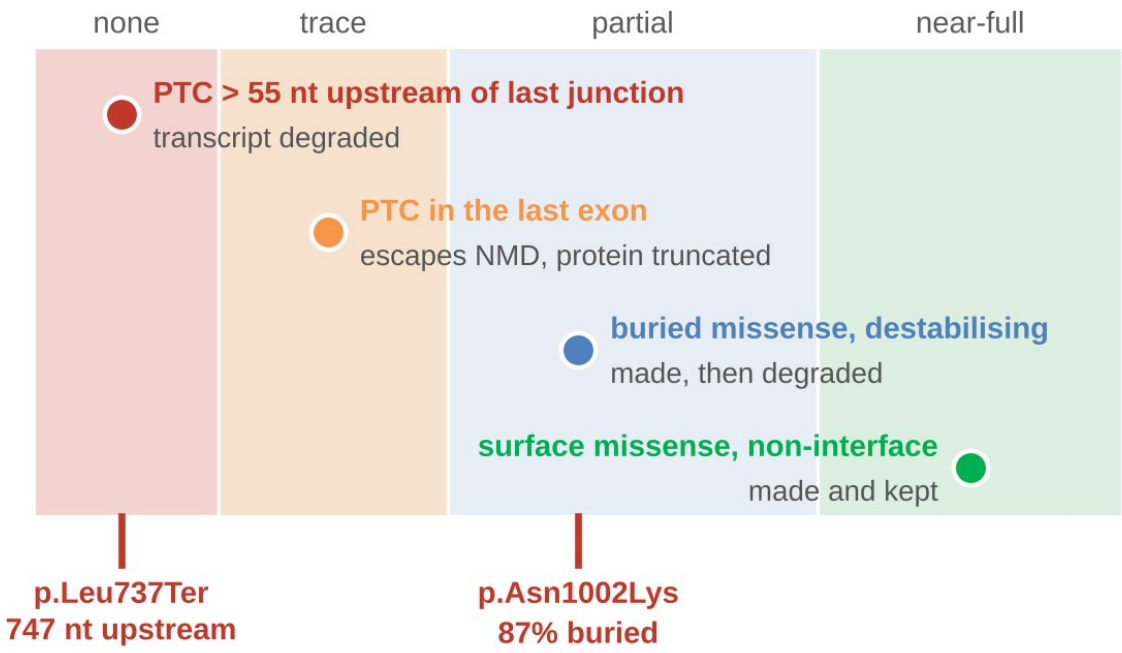
consequence call
truncating / missense / splice

NMD rule, 55 nt
does the transcript survive

AlphaMissense + GPN-Star
two predictors, one adds splice

literature abundance
no predictor has this axis

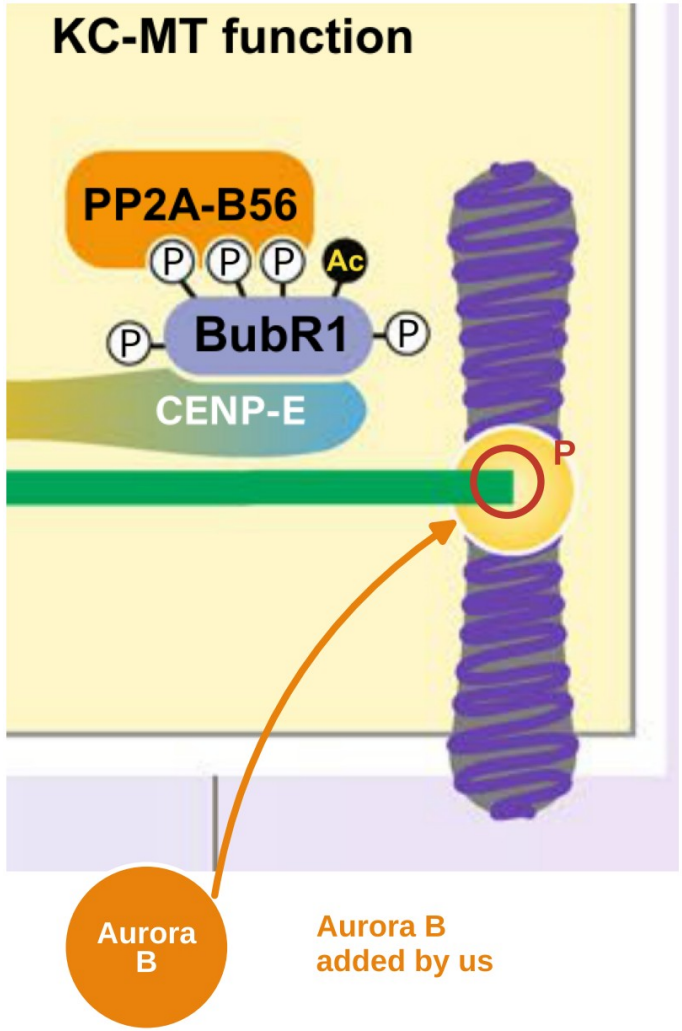
Expected BubR1 output, by allele class



Suijkerbuijk reached the same two classes experimentally: truncating alleles lose the transcript, missense alleles are made and turned over.

Suijkerbuijk et al., Cancer Res 2010, PMID 20516114

Q1 — Aurora B LOOSENS grips on purpose. PP2A-B56 SETS them. This child cannot set.



AURORA B — the loosener

It phosphorylates the GRIP (Hec1) holding a chromosome to a microtubule. A phosphorylated grip lets go. That is how wrong attachments get broken.

PP2A-B56 — the setter

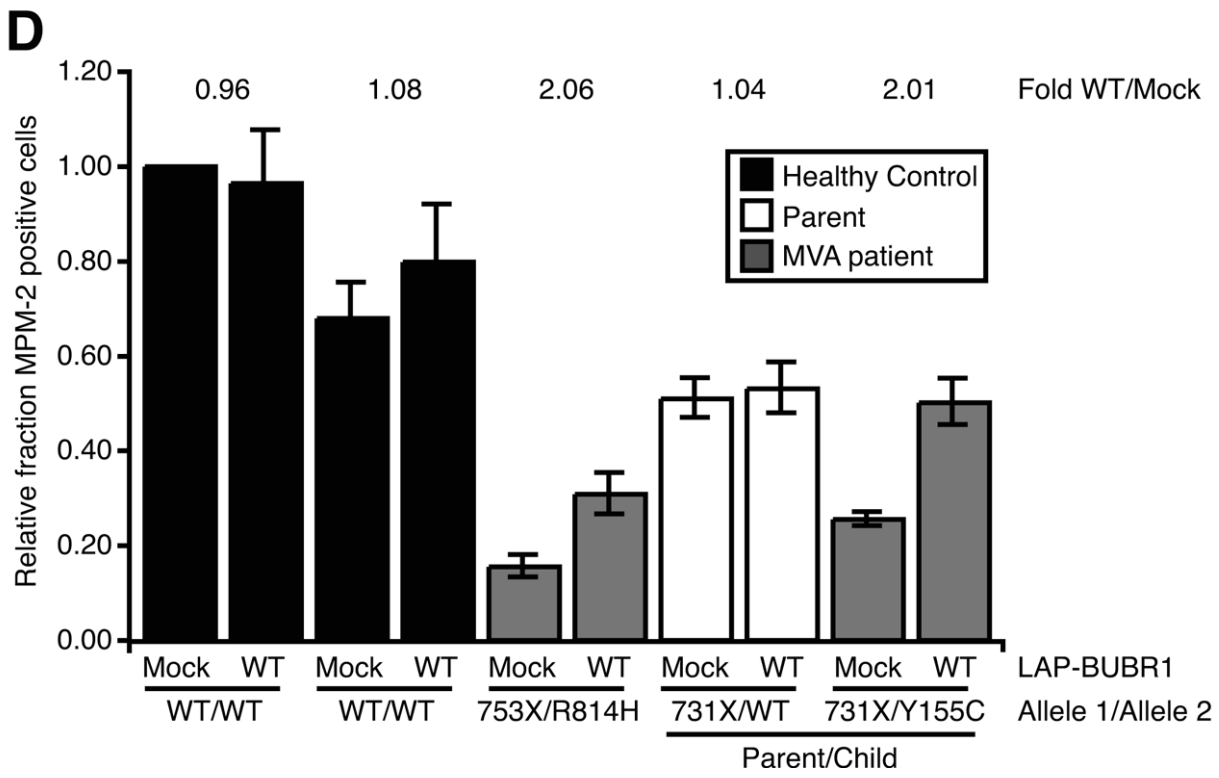
It removes those phosphates so a correct grip can set. Error correction is these two in a tug-of-war, and both are needed.

BubR1 is the TAXI, not an enzyme. Plk1 puts 3 phosphates in its KARD (shown left, on BubR1): that is the TICKET PP2A-B56 binds. Different phosphates, opposite job to Aurora B's.

No KARD delivered → no setter arrives → Aurora B is not stronger, it is UNOPPOSED. So we lower it to match.

Bloom & North 2021, Fig 2 detail; Aurora B added by us

Q2 — The deficit is reversible: put BubR1 back and the checkpoint recovers



This child's genotype shape

753X/R814H and 731X/Y155C: one truncating allele, one missense — the MVA pattern.

2.06x and 2.01x

Nocodazole arrest doubles when BubR1 is restored. Healthy controls and the carrier parent do not move.

But it does not normalise

15% → 31% and 26% → 50% of healthy. And no drug raises BubR1.

Suijkerbuijk et al., Cancer Res 2010, PMID 20516114, Fig 1D

SCIENTIFIC RIGOR (35%) — Q3: ten arms, four themes, every one closed on a number

Scientific rigor 35%

Rescue allele 1
the truncated one

- ~~Translational readthrough~~
- ~~NMD inhibition~~
- ~~Gene / ASO / trans-splicing~~

closed on exposure

Rescue allele 2
the missense one

- ~~Chaperone, HSP90~~
- ~~Degradation, SIRT2 / NAD+~~
- ~~Proteasome inhibition~~

closed on exposure

Target BubR1 itself
make it, or bind it

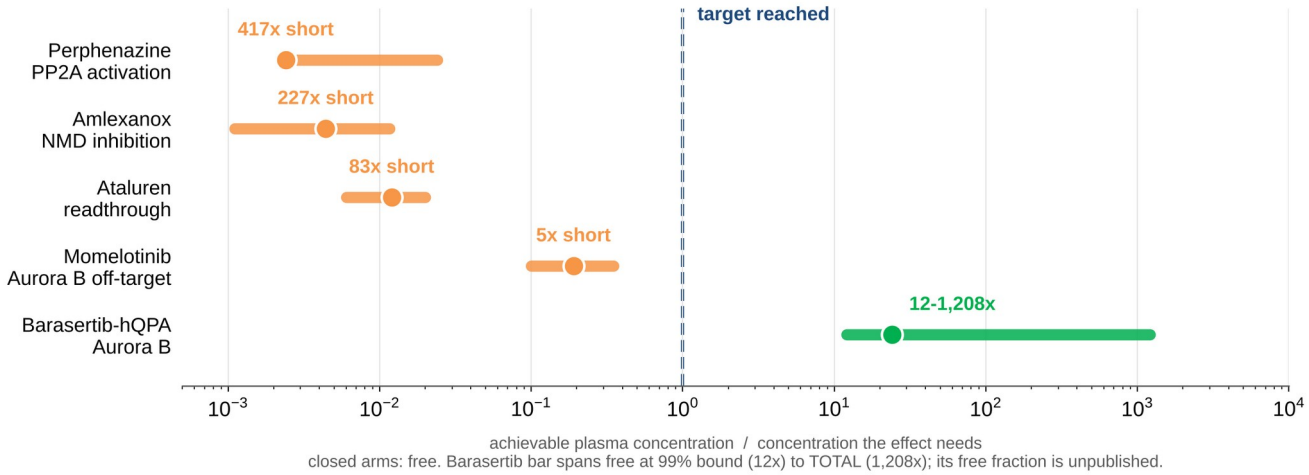
- ~~Direct small-molecule binding~~

closed on exposure

Rebalance downstream
where both alleles meet

- ~~Activate PP2A directly~~
- ~~Restrain APC/C~~
- **Lower Aurora B**

SURVIVES

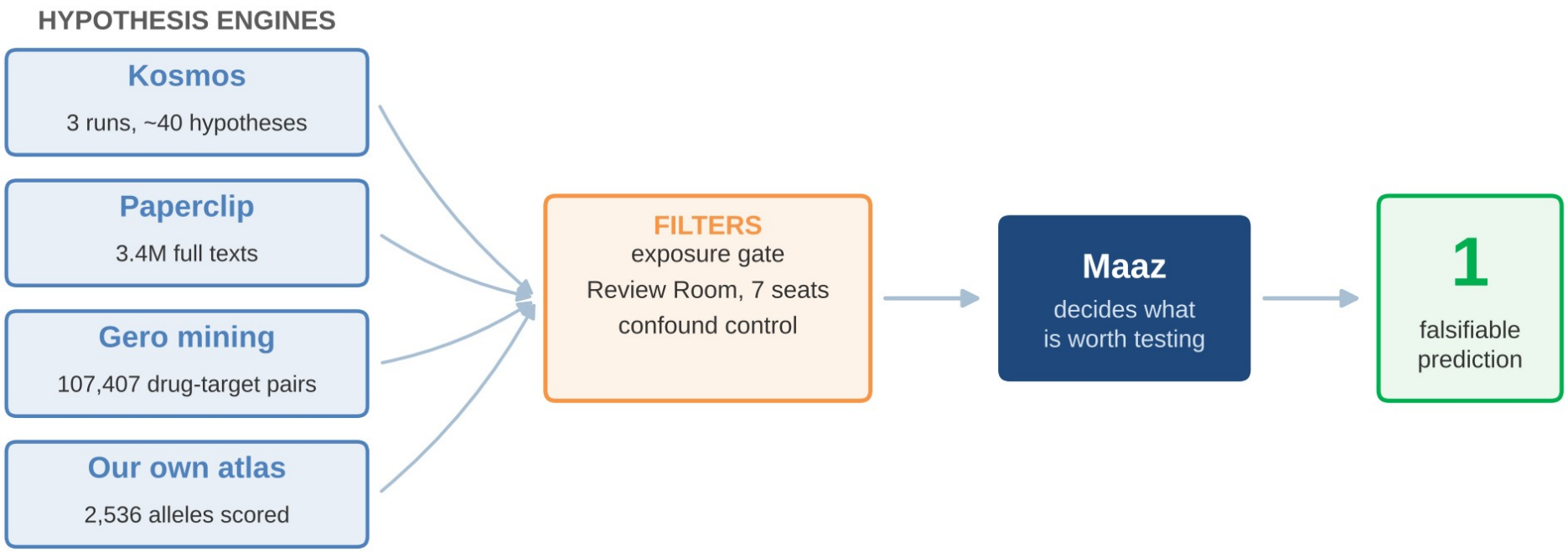


Two gates closed these arms. EXPOSURE: most needed more drug than a person can be given. SAFETY IN THIS CHILD: proteasome and HSP90 inhibitors are myelosuppressive and carcinogenic, disqualifying where tumour risk already comes from missegregation. Barasertib alone sits ABOVE what the effect needs, by 12x on the most conservative free-drug assumption.

Three themes die upstream. One node survives.

INNOVATION (25%) — many engines generate, one human decides, nothing rests on one paper

Innovation 25%

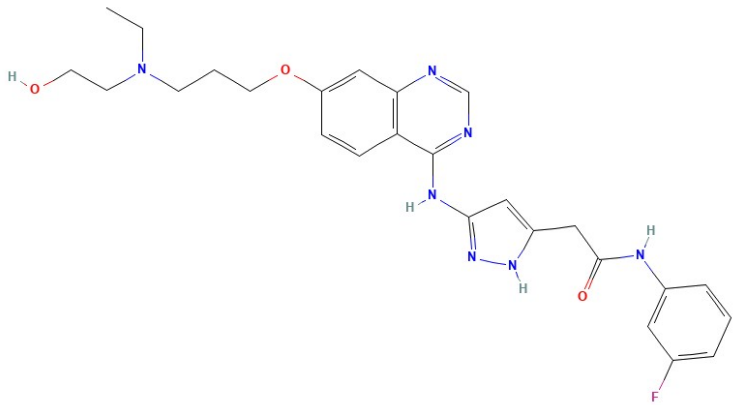


~40 hypotheses → 10 arms survived scrutiny → 1 committed prediction

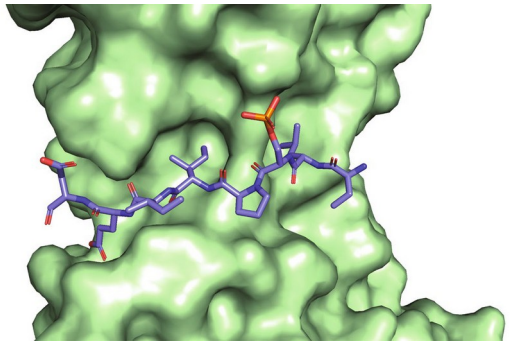
54	5	4	2,536	51	0
primary papers read, not cited	drug-target databases	experimental structures	alleles scored in our atlas	numbered corrections to our own work	claims resting on a single paper

POTENTIAL IMPACT (25%) — Q4: lower the kinase, because the phosphatase cannot be raised

Potential impact 25%



barasertib-hQPA



the groove neither allele reaches

0.37 nM BIOCHEMICAL Aurora B IC50, not the cellular rescue concentration

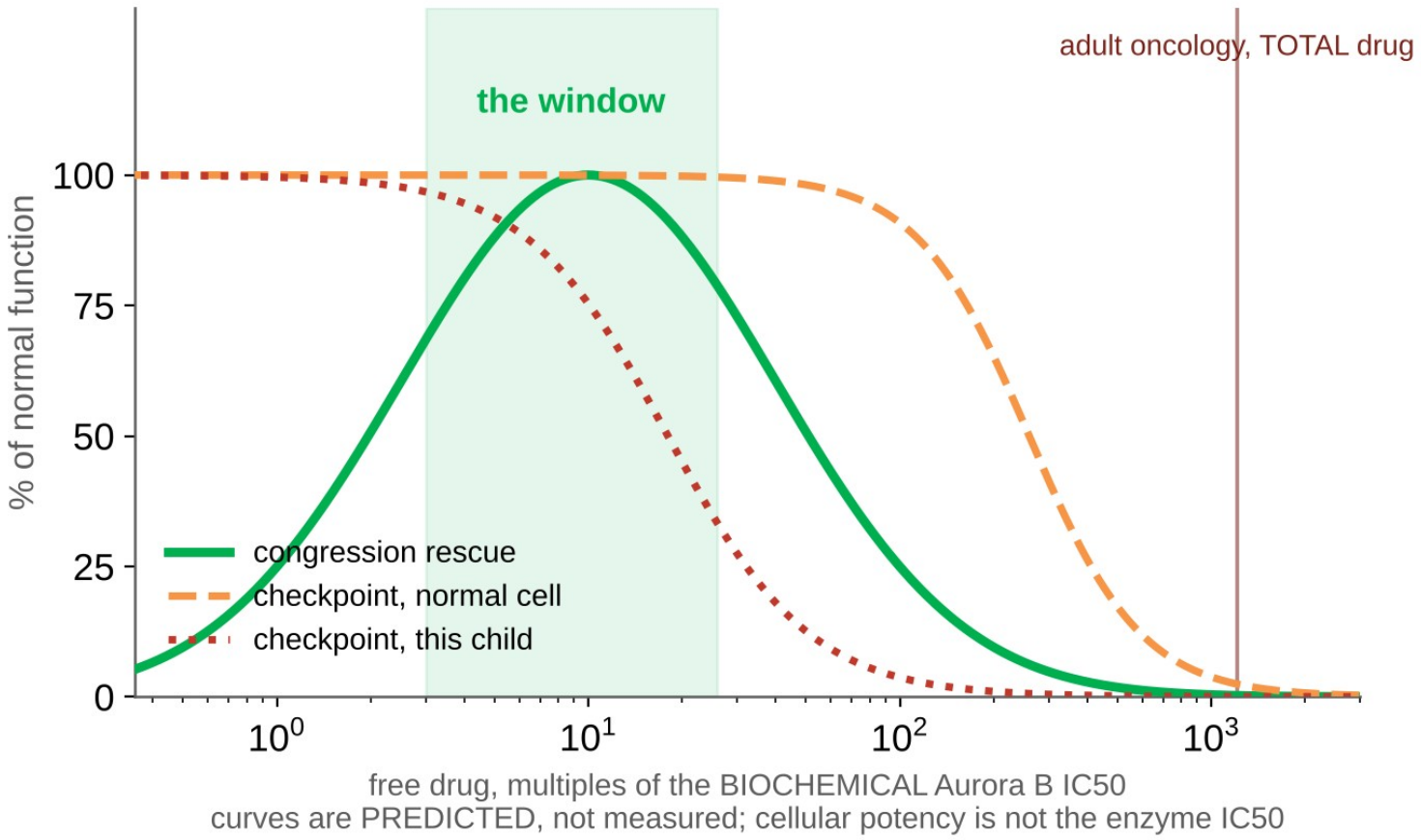
~3,700x selective over Aurora A

447 nM adult steady-state Css, TOTAL drug (Dennis 2012)

12-1,208x over IC50: free at 99% bound, up to total. Free fraction unpublished.

A RANGE, not a therapeutic index and not a safety margin: it shows substantial exposure is reachable, not that a safe paediatric window exists. Structure: PP2A-B56 with the BubR1 KARD bound (Bloom & North 2021).

The dose IS the hypothesis: set by the cells, not asserted



Xu used ZM447439 — equipotent on Aurora A and B. Barasertib is the selective successor. The mechanism transfers; the numbers do not.

1-5 nM
free drug, the target

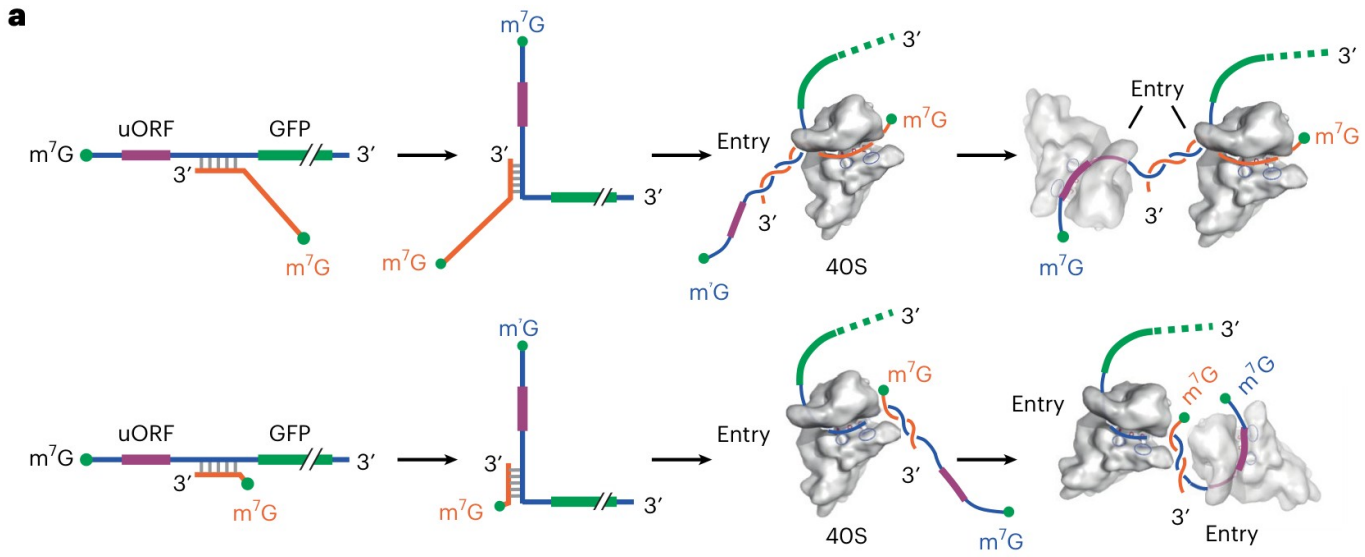
30-50%
loss of phospho-Hec1.
Never ablation.

0.1-30 nM
titrated in the child's
own fibroblasts

Score MISSEGREGATION,
not alignment

Ditchfield 2003 · Xu 2013 · Santaguida 2011

The moonshot: programmable translation, on the message the patient already has



What it does

A separate capped RNA (orange) base-pairs with the target mRNA (blue). The ribosome loads from the trans-cap and is handed to a start codon of our choosing.

Why it reaches this child

Allele 1's transcript is gone, so nothing can rescue it. But allele 2's message IS present — raising its output is a genome-free way to restore the dose.

Why it deserves funding

One mechanism, no edit, and dosage disease is a whole class. Delivery is the unsolved half: liver and muscle today, brain and marrow not yet.

Jia et al., Nat Biotechnol 2025, Fig 1a

What we answered, and the one prediction we commit to

Q1 What do the alleles leave? No transcript, and an unstable protein. Both lose the KARD's function.

Q2 Is it reversible? Yes — restoring BubR1 doubles checkpoint arrest in patient cells.

Q3 Can a drug raise BubR1? No. Zero compounds across 107,407 curated pairs, and no PP2A activator.

Q4 What else sits on the node? Aurora B — the one target where human exposure runs in our favour.

The prediction: partial Aurora B inhibition rescues congression in this child's cells BEFORE it costs checkpoint competence. If it does not, we are wrong.